Riemann-Roch Theorem on $\mathbb{CP}^1$

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We cover the following topics in this note.

- Complex Projective Line $\mathbb{CP}^1$
- (Principal) Divisors on $\mathbb{CP}^1$
- Riemann-Roch Space $L(D)$
- TBA

On a general compact Riemann surface X, **Riemann–Roch** says

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where

- D is a divisor,
- K is a canonical divisor,
- g is the genus,
- $\ell(D) = \dim_{\mathbb{C}} L(D)$, where

$$L(D) := \{f \in \mathcal{M}(X) : (f) + D \geq 0\} \cup \{0\}.$$

For $X = \mathbb{CP}^1$ we have $g = 0$, so the theorem becomes

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

The key observation on $\mathbb{CP}^1$ is that every divisor is linearly equivalent to a multiple of $[\infty]$:

$$D \sim (\deg D) \cdot [\infty].$$

Contents

1 Coordinates on $\mathbb{CP}^1$ **3**

2 Divisors of Meromorphic Functions **4**

3 Every Divisor on $\mathbb{CP}^1$ is Equivalent to $n[\infty]$ **9**

4 The Space of Meromorphic Functions $L(D)$ **10**

5 The Canonical Divisor **37**

6 Riemann–Roch Theorem **37**

Notation

$\mathcal{O}(X)$ Field of holomorphic functions on X

$\mathcal{O}(X)^\times$ Field of holomorphic functions on X except vanishing function

$\mathcal{M}(X)$ Field of meromorphic functions on X

$\mathcal{M}(X)^\times$ Field of meromorphic functions on X except zero function

$\mathcal{O}_X^\times$ Sheaf of holomorphic functions on X

$\mathcal{M}_X^\times$ Sheaf of meromorphic functions on X

$L(D)$ The vector space of meromorphic functions whose poles are bounded by D

$\ell(D)$ $\dim_{\mathbb{C}} L(D)$

1 Coordinates on $\mathbb{CP}^1$

Complex Projective Line

Definition 1. The **complex projective line** is the quotient

$$\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{(0,0)\})/\sim,$$

where

$$(Z_0, Z_1) \sim (Z_0', Z_1') \iff (Z_0, Z_1) = (\lambda Z_0', \lambda Z_1')$$

for some $\lambda \in \mathbb{C}^\times$. The class of (Z_0, Z_1) is denoted by $[Z_0 : Z_1]$.

Remark 1. The two standard affine charts are

$$U_0 = \{[Z_0 : Z_1] \in \mathbb{CP}^1 : Z_0 \neq 0\}, \qquad U_\infty = \{[Z_0 : Z_1] \in \mathbb{CP}^1 : Z_1 \neq 0\}.$$

On U_0 one uses the coordinate $z = \frac{Z_1}{Z_0}$, so

$$U_0 = \{[Z_0 : Z_1] \in \mathbb{CP}^1 : Z_0 \neq 0\} = \{[1 : z] \in \mathbb{CP}^1 : z \in \mathbb{C}\} \simeq \mathbb{C}.$$

On U_∞ one uses $w = \frac{Z_0}{Z_1}$, so

$$U_\infty = \{[Z_0 : Z_1] \in \mathbb{CP}^1 : Z_1 \neq 0\} = \{[w : 1] \in \mathbb{CP}^1 : w \in \mathbb{C}\} \simeq \mathbb{C}.$$

On the overlap $U_0 \cap U_\infty$ these coordinates satisfy $w = \frac{1}{z}$.

Remark 2. The distinguished point

$$\infty = [0 : 1]$$

lies in U_∞ and corresponds to $w = 0$. In the chart $U_0 \cong \mathbb{C}$, the projective line looks like the complex plane together with one additional point at infinity.

2 Divisors of Meromorphic Functions

Divisor; Degree of Divisor

Definition 2. A **divisor** on $\mathbb{CP}^1$ is a finite formal sum

$$D = \sum_{p \in \mathbb{CP}^1} n_p[p], \qquad n_p \in \mathbb{Z}.$$

Its **degree** is $\deg D = \sum_{p \in \mathbb{CP}^1} n_p$.

Remark 3. If $f \in \mathcal{M}(\mathbb{CP}^1)^\times$ and $p \in X$, then

$$\mathrm{ord}_p(f) = \mathrm{Res}_p\left(\mathrm{d}(\log f)\right) = \mathrm{Res}_p\left(\frac{\mathrm{d}f}{f}\right) = \frac{1}{2\pi i}\oint_{\gamma_p} \frac{\mathrm{d}f}{f}.$$

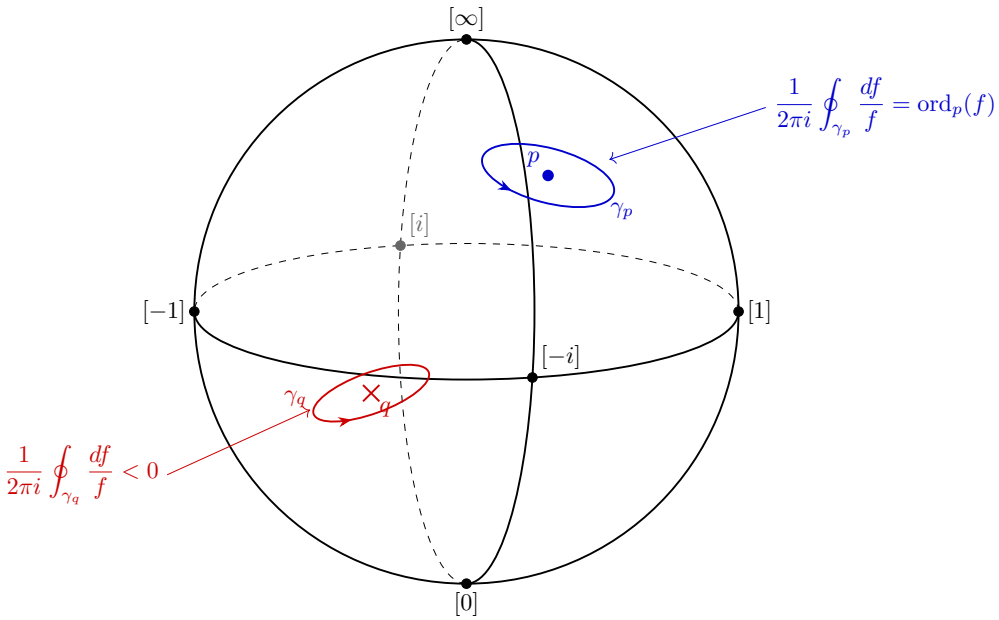
The **principal divisor** of f is

$$\mathrm{div}(f) = \sum_{p \in \mathbb{CP}^1} \mathrm{ord}_p(f)[p] = \sum_{p \in \mathbb{CP}^1} \left(\frac{1}{2\pi i}\oint_{\gamma_p} \frac{\mathrm{d}f}{f}\right)[p].$$

Here,

$$\deg(\mathrm{div}(f)) = \sum_{p \in \mathbb{CP}^1} \mathrm{ord}_p(f) = \sum_{p \in \mathbb{CP}^1} \left(\frac{1}{2\pi i}\oint_{\gamma_p} \frac{\mathrm{d}f}{f}\right).$$

Remark 4. We also write (f) for $\mathrm{div}(f)$.



Example 1. In the coordinate z on U_0, the function $z - a$ has a simple zero at $a \in \mathbb{C}$. To understand what happens at ∞, write $z = 1/w$. Then

$$z - a = \frac{1 - aw}{w}.$$

Near $w = 0$, the factor $1 - aw$ is a unit, while $1/w$ has a simple pole. Therefore

$$(z - a) = [a] - [\infty].$$

Example 2 (The divisor of $z-a$ on $\mathbb{CP}^1$)**.** Now fix $a \in \mathbb{C}$. On the chart U_0, the meromorphic function $z - a$ has a simple zero at the point $[1 : a]$, since $z = a$ there.

To determine its behavior at ∞, pass to the chart U_∞. Since

$$z = \frac{1}{w},$$

we get

$$z - a = \frac{1}{w} - a = \frac{1 - aw}{w}.$$

Near $w = 0$, the factor $1 - aw$ is holomorphic and nonvanishing, hence a unit. Therefore the order of $z - a$ at ∞ is the same as the order of $1/w$, namely

$$\mathrm{ord}_\infty(z - a) = -1.$$

Thus $z - a$ has a simple zero at $[1 : a]$ and a simple pole at ∞. Hence

$$\mathrm{div}(z - a) = [a] - [\infty].$$

Example 3. More generally,

$$\left(\frac{z - a}{z - b}\right) = [a] - [b].$$

Lemma 1. *Let X be a compact Riemann surface, and let $f \in \mathcal{M}(X)^\times$ be a nonzero meromorphic function. Then*

$$\sum_{p \in X} \operatorname{ord}_p(f) = 0.$$

Proof. Let $p_1, \dots, p_N$ are the finitely many zeros and poles of f.

Choose pairwise disjoint coordinate disks U_j around p_j, with local coordinate $z_j : U_j \to \Delta$ satisfying $z_j(p_j) = 0$. For sufficiently small $r > 0$, let

$$D_j(r) := \{q \in U_j : |z_j(q)| < r\}, \qquad C_j(r) := \partial D_j(r).$$

Set

$$X_r := X \setminus \bigcup_{j=1}^{N} D_j(r).$$

Then X_r is a compact oriented surface with boundary

$$\partial X_r = -\sum_{j=1}^{N} C_j(r),$$

because each $C_j(r)$ appears with the boundary orientation induced from the complement of the deleted disk.

On X_r, the function f has no zeros and no poles, so the 1-form

$$\omega := \frac{df}{f}$$

is smooth on X_r. Since locally $\omega = d(\log f)$ on X_r, we have

$$d\omega = 0.$$

Therefore, by the generalized Stokes theorem,

$$0 = \int_{X_r} d\omega = \int_{\partial X_r} \omega = -\sum_{j=1}^{N} \int_{C_j(r)} \frac{df}{f}.$$

Hence

$$\sum_{j=1}^{N} \int_{C_j(r)} \frac{df}{f} = 0.$$

Now compute each boundary integral locally. Since $m_j = \operatorname{ord}_{p_j}(f)$, in the coordinate z_j we may

write

$$f = z_j^{m_j} u_j,$$

where u_j is holomorphic and nowhere vanishing on U_j. Thus

$$\frac{df}{f} = m_j \frac{dz_j}{z_j} + \frac{du_j}{u_j}.$$

Integrating around $C_j(r)$ gives

$$\int_{C_j(r)} \frac{df}{f} = m_j \int_{C_j(r)} \frac{dz_j}{z_j} + \int_{C_j(r)} \frac{du_j}{u_j}.$$

Because u_j is nowhere vanishing on U_j, there exists a holomorphic branch of $\log u_j$ on U_j, so

$$\frac{du_j}{u_j} = d(\log u_j),$$

and therefore

$$\int_{C_j(r)} \frac{du_j}{u_j} = 0.$$

Also,

$$\int_{C_j(r)} \frac{dz_j}{z_j} = 2\pi i.$$

So

$$\int_{C_j(r)} \frac{df}{f} = 2\pi i \, m_j.$$

Substituting into the global identity, we obtain

$$0 = \sum_{j=1}^{N} \int_{C_j(r)} \frac{df}{f} = 2\pi i \sum_{j=1}^{N} m_j.$$

Hence

$$\sum_{j=1}^{N} m_j = 0,$$

that is,

$$\sum_{p \in X} \operatorname{ord}_p(f) = 0.$$

This proves that every principal divisor on a compact Riemann surface has degree zero. □

Proposition 2. *Every principal divisor on $\mathbb{CP}^1$ has degree 0.*

Proof. Let $f = P/Q$ with coprime polynomials. If $\deg P = m$ and $\deg Q = n$, then the finite zeros contribute m, the finite poles contribute $-n$, and near infinity we have

$$f(1/w) = \frac{P(1/w)}{Q(1/w)} \sim c\, w^{n-m},$$

so

$$\operatorname{ord}_\infty(f) = n - m.$$

Hence

$$\deg(f) = m - n + (n - m) = 0.$$

□

3 Every Divisor on $\mathbb{CP}^1$ is Equivalent to $n[\infty]$

Linearly Equivalence

Definition 3. Two divisors D, E are linearly equivalent, written $D \sim E$, if

$$D - E = (f)$$

for some nonzero meromorphic function f.

The key fact on $\mathbb{CP}^1$ is that degree is the only invariant of divisor class.

Theorem 3. *Every divisor D on $\mathbb{CP}^1$ is linearly equivalent to $(\deg D)[\infty]$. In particular,*

$$\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}, \qquad D \longmapsto \deg D.$$

Proof. Write

$$D = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j] + c[\infty],$$

where $a_i, b_j \in \mathbb{C}$ and $m_i, n_j > 0$. Define

$$h(z) = \frac{\prod_{i=1}^{r}(z-a_i)^{m_i}}{\prod_{j=1}^{s}(z-b_j)^{n_j}}.$$

Then

$$(h) = \sum_{i=1}^{r} m_i[a_i] - \sum_{j=1}^{s} n_j[b_j] - \left(\sum_{i=1}^{r} m_i - \sum_{j=1}^{s} n_j\right)[\infty].$$

Therefore

$$D - (h) = \left(c + \sum_{i=1}^{r} m_i - \sum_{j=1}^{s} n_j\right)[\infty] = (\deg D)[\infty].$$

So $D \sim (\deg D)[\infty]$. □

This theorem is the concrete genus-zero form of the statement that every line bundle on $\mathbb{CP}^1$ is some $\mathcal{O}(n)$.

4 The Space of Meromorphic Functions $L(D)$

Once divisors encode the zeros and poles of meromorphic functions, a natural question is

Which meromorphic functions have zeros and poles compatible with a given divisor?

This leads to the definition

$$L(D) = \{ f \in \mathcal{M}(X) \mid \operatorname{div}(f) + D \geq 0 \} \cup \{0\},$$

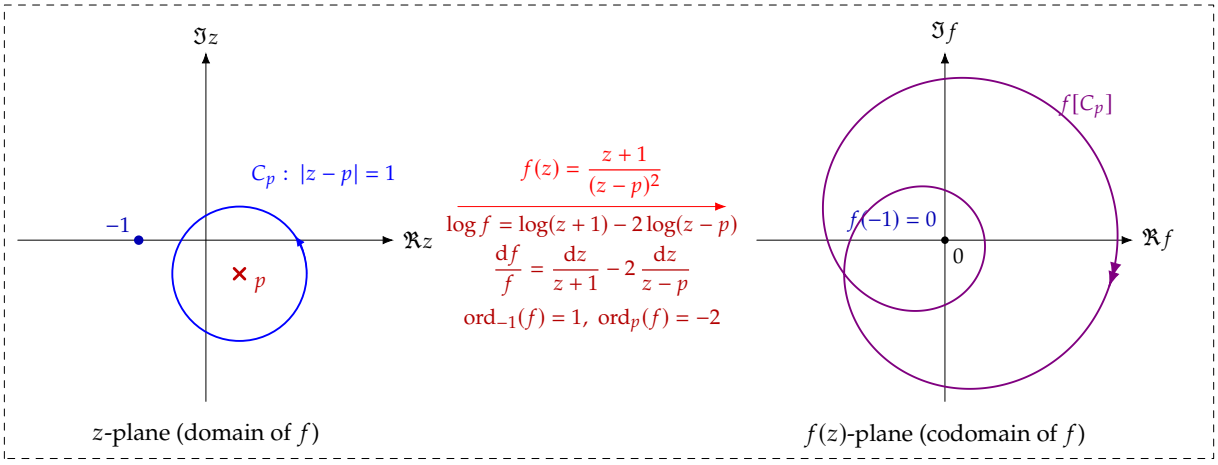
the space of meromorphic functions whose poles are bounded by D.

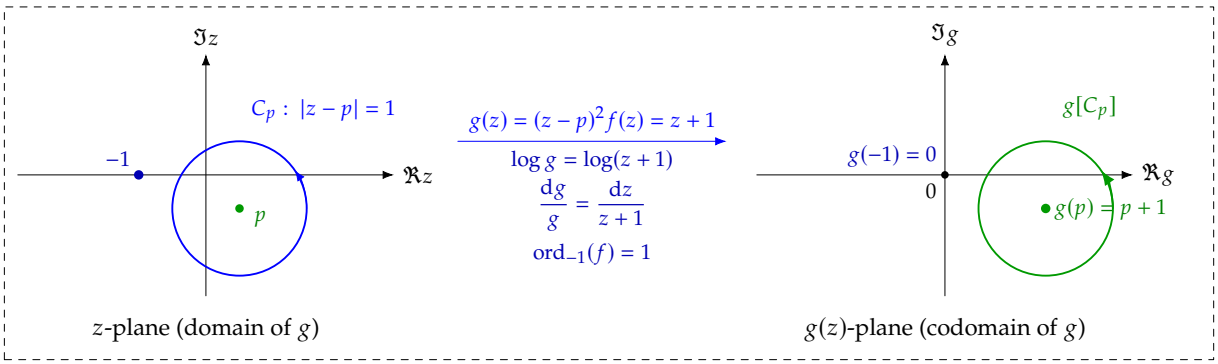
Observation. Let $p \in X$, and let z be a local coordinate with $z(p) = 0$. Any nonzero meromorphic function f near p can be written as

$$f = z^{-m} u, \quad u(p) \neq 0,$$

for some $m \in \mathbb{Z}$. Now suppose we want to allow at most a pole of order n at p. Then

$$f \text{ has pole of order at most } n \text{ at } p \iff z^n f \text{ is holomorphic at } p.$$





Thus

$$\begin{aligned}\operatorname{ord}_p(z^n f) = \frac{1}{2\pi i}\oint_{\gamma_p}\frac{\mathrm{d}(z^n f)}{z^n f} &= \frac{1}{2\pi i}\oint_{\gamma_p}\left(\frac{f\cdot nz^{n-1}\mathrm{d}z + z^n\cdot f'\mathrm{d}z}{z^n f}\right)\\ &= \frac{1}{2\pi i}\oint_{\gamma_p}\left(n\cdot\frac{\mathrm{d}z}{z}+\frac{\mathrm{d}f}{f}\right)\\ &= \frac{n}{2\pi i}\oint_{\gamma_p}\frac{\mathrm{d}z}{z}+\frac{1}{2\pi i}\oint_{\gamma_p}\frac{\mathrm{d}f}{f}\\ &= n\cdot\frac{2\pi i}{2\pi i}+\operatorname{ord}_p(f)\\ &= n+\operatorname{ord}_p(f)\geq 0,\end{aligned}$$

so

$$f \text{ has pole order at most } n \text{ at } p \iff z^n f \text{ is holomorphic at } p \iff \operatorname{ord}_p(f)+n\geq 0.$$

Divisor Group

Definition 4. Let X be a compact Riemann surface. The **divisor group** is

$$\operatorname{Div}(X) := \left\{\sum_{p\in X} n_p[p] \;\middle|\; n_p\in\mathbb{Z}, \text{ and } n_p = 0 \text{ for all but finitely many } p\right\}.$$

The group operation is addition:

$$\left(\sum_p n_p[p]\right)+\left(\sum_p m_p[p]\right)=\sum_p (n_p+m_p)[p].$$

Since addition of integers is commutative, $\operatorname{Div}(X)$ is an abelian group.

Remark 5. Consider the following isomorphism

$$\begin{array}{rccc}\Phi & : & \operatorname{Div}(X) & \longrightarrow & \bigoplus_{p\in X}\mathbb{Z}\\ & & \sum_{p\in X} n_p[p] & \longmapsto & (n_p)_{p\in X}\end{array}.$$

That is, $\operatorname{Div}(X)\cong\bigoplus_{p\in X}\mathbb{Z}$, so $\operatorname{Div}(X)$ is the free abelian group generated by the points of X.

Remark 6. A divisor $D=\sum_p n_p[p]$ is called **effective** if $n_p\geq 0$ for all $p\in X$, and we write $D\geq 0$.

Space of Meromorphic Functions (Riemann–Roch Space)

Definition 5. Let

$$D = \sum_{p \in \mathbb{CP}^1} n_p[p] \in \mathrm{Div}(\mathbb{CP}^1)$$

be a divisor on $\mathbb{CP}^1$. The **space of meromorphic functions associated to** D is

$$L(D) := \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + D \geq 0 \} \cup \{0\}.$$

Equivalently,

$$L(D) = \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_p(f) \geq -n_p \text{ for every } p \in \mathbb{CP}^1 \} \cup \{0\}.$$

Its dimension is denoted by

$$\ell(D) := \dim_{\mathbb{C}} L(D).$$

Remark 7. For a general divisor $D = \sum_{p \in X} n_p[p]$ on a compact Riemann surface X,

$$f \in L(D) \iff \mathrm{ord}_p(f) + n_p \geq 0 \text{ for all } p \in X.$$

Therefore

- if $n_p > 0$, then f may have a pole at p, but of order at most n_p;
- if $n_p = 0$, then f cannot have a pole at p;
- if $n_p < 0$, then f must have a zero at p of order at least $-n_p$.

Remark 8. Take $D = \sum_{p \in X} n_p[p]$ and consider $L(D)$. Then

(i) By definition, $0 \in L(D)$.

(ii) Let $f \in L(D)$ and $c \in \mathbb{C}$. If $c = 0$, then $cf = 0 \in L(D)$. If $c \neq 0$, then

$$\mathrm{ord}_p(cf) = \frac{1}{2\pi i} \oint_{\gamma_p} \frac{\mathrm{d}(cf)}{cf} = \frac{1}{2\pi i} \oint_{\gamma_p} \frac{\mathrm{d}f}{f} = \mathrm{ord}_p(f).$$

for every point $p \in X$. Then

$$\mathrm{ord}_p(f) + n_p \geq 0 \implies \mathrm{ord}_p(cf) + n_p \geq 0$$

for all $p \in X$, hence $cf \in L(D)$.

(iii) Let $f, g \in L(D)$ with $f + g \neq 0$. Fix a point $p \in X$. Define

$$m := \min\{\mathrm{ord}_p(f), \mathrm{ord}_p(g)\}.$$

Choose a local coordinate z centered at p. Then

$$f = z^m u, \qquad g = z^m v,$$

where u and v are holomorphic near p, and at least one of $u(p), v(p)$ is nonzero. Then

$$\mathrm{ord}_p(f + g) = \mathrm{ord}_p(z^m(u + v)) = m + \mathrm{ord}_p(u + v).$$

Since $u + v$ is holomorphic ($\mathrm{ord}_p(u + v) \geq 0$), we have

$$\mathrm{ord}_p(f + g) \geq m = \min\{\mathrm{ord}_p(f), \mathrm{ord}_p(g)\}.$$

Since $f, g \in L(D)$, we have

$$\mathrm{ord}_p(f) + n_p \geq 0 \quad \text{and} \quad \mathrm{ord}_p(g) + n_p \geq 0,$$

so

$$m + n_p = \min\{\mathrm{ord}_p(f), \mathrm{ord}_p(g)\} + n_p \geq 0.$$

Therefore

$$\mathrm{ord}_p(f + g) + n_p \geq m + n_p \geq 0,$$

so $f + g \in L(D)$.

Hence $L(D)$ is a vector space over $\mathbb{C}$. If X is compact, this vector space is in fact finite-dimensional, and we write

$$\ell(D) := \dim_{\mathbb{C}} L(D).$$

Example 4 (Computing $L(D)$ and $\ell(D)$)**.**

(1) Let $D = 0$ (zero divisor). Then

$$L(0) = \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) \geq 0 \,\} \cup \{0\} = \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_p(f) \geq 0 \text{ for all } p \in \mathbb{CP}^1 \} \cup \{0\}.$$

So f has no poles anywhere on $\mathbb{CP}^1$. Hence f is holomorphic everywhere on the compact Riemann surface $\mathbb{CP}^1$, so f must be constant. Therefore

$$L(0) = \mathbb{C} \quad \text{and} \quad \ell(0) = \dim_{\mathbb{C}} L(0) = 1.$$

(2) Let $D = [0]$ (simple zero at 0). Then

$$\begin{aligned} L([0]) &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [0] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f)+1)[0] + (\operatorname{ord}_\infty(f))[\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \notin \{0,\infty\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq -1;\ \operatorname{ord}_\infty(f) \geq 0 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has no pole at } \infty \\ f \text{ may have pole of order} \leq 1 \text{ at } 0 \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = \frac{P(z)}{z}, \quad P(z) \in \mathbb{C}[z].$$

Note that

$$\begin{aligned} P(z) &= a_0 + a_1 z + \cdots + a_{\deg P} z^{\deg P} \quad (a_{\deg P} \neq 0), \\ P(1/w) &= a_0 + a_1 w^{-1} + \cdots + a_{\deg P} w^{-\deg P}. \end{aligned}$$

Since no pole is allowed at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}(wP(1/w)) = 1 - \deg P \geq 0,$$

so $\deg P \leq 1$. Thus $P(z) = az + b$ $(a, b \in \mathbb{C})$ and therefore

$$f(z) = \frac{az+b}{z} = a + \frac{b}{z}.$$

Hence

$$L([0]) = \left\{ a + \frac{b}{z} : a, b \in \mathbb{C} \right\} = \operatorname{span}_{\mathbb{C}} \left\{ 1, \frac{1}{z} \right\} \quad \text{so} \quad \ell([0]) = 2.$$

(3) Let $D = [0] + [1]$. Then

$$\begin{aligned} &L([0]+[1]) \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [0] + [1] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0,1\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f)+1)[0] + (\operatorname{ord}_1(f)+1)[1] + (\operatorname{ord}_\infty(f))[\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \notin \{0,\infty\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq -1;\ \operatorname{ord}_1(f) \geq -1;\ \operatorname{ord}_\infty(f) \geq 0 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has no pole at } \infty \\ f \text{ may have pole of order} \leq 1 \text{ at } 0 \\ f \text{ may have pole of order} \leq 1 \text{ at } 1 \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = \frac{P(z)}{z(z-1)}, \quad P(z) \in \mathbb{C}[z].$$

Since no pole is allowed at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(w \cdot \frac{w}{1-w} \cdot P(1/w)\right) = \mathrm{ord}_{w=0}\left(w^2 \cdot \frac{1}{1-w} \cdot P(1/w)\right) = 2 - \deg P \geq 0,$$

so $\deg P \leq 2$. Thus $P(z) = az^2 + bz + c$ $(a, b, c \in \mathbb{C})$ and therefore

$$f(z) = \frac{az^2 + bz + c}{z(z-1)} = A + \frac{B}{z} + \frac{C}{z-1}.$$

Since

$$az^2 + bz + c = Az(z-1) + B(z-1) + Cz = Az^2 + (-A + B + C)z - B,$$

we have

$$\begin{cases} A = a \\ -A + B + C = b \\ -B = c \end{cases} \implies \begin{cases} A = a \\ B = -c \\ C = a + b + c \end{cases}.$$

Thus

$$f(z) = a - \frac{c}{z} + \frac{a+b+c}{z-1}, \qquad a, b, c \in \mathbb{C}.$$

Therefore

$$L([0] + [1]) = \left\{ a - \frac{c}{z} + \frac{a+b+c}{z-1} : a, b, c \in \mathbb{C} \right\} = \mathrm{span}\left\{1, \frac{1}{z}, \frac{1}{z-1}\right\} \quad \text{and} \quad \ell([0] + [1]) = 3.$$

(4) Let $D = [\infty]$. Then

$$\begin{aligned} L([\infty]) &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + [\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}} \mathrm{ord}_p[p] + (\mathrm{ord}_\infty(f) + 1)[\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p \in \mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -1 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{matrix} f \text{ may have a pole of order} \leq 1 \text{ at } \infty \\ f \text{ has no pole at any finite point } p \in \mathbb{C} \end{matrix} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 1 at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(P(1/w)\right) = -\deg P \geq -1,$$

so $\deg P \leq 1$. Thus $P(z) = az + b$ $(a, b \in \mathbb{C})$ and therefore

$$f(z) = az + b, \quad a, b \in \mathbb{C}.$$

Hence

$$L([\infty]) = \{az + b : a, b \in \mathbb{C}\} = \mathrm{span}_{\mathbb{C}}\{1, z\} \quad \text{and} \quad \ell([\infty]) = 2.$$

(5) Let $D = 2[\infty]$. Then

$$L(2[\infty]) = \{\, f \in \mathcal{M}(\mathbb{CP}^1) \mid \mathrm{div}(f) + 2[\infty] \geq 0 \,\} \cup \{0\}.$$

Thus f has no poles on $\mathbb{C}$ and at most a double pole at ∞. Equivalently, f is a polynomial of degree at most 2. Therefore

$$L(2[\infty]) = \left\{az^2 + bz + c : a, b, c \in \mathbb{C}\right\} = \mathrm{span}_{\mathbb{C}}\{1, z, z^2\} \quad \text{and} \quad \ell(2[\infty]) = 3.$$

(6) Let $D = [0] + [\infty]$. Then

$$\begin{aligned} L([0]+[\infty]) &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [0] + [\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p\in\mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f)+1)[0] + (\operatorname{ord}_\infty(f)+1)[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p\in\mathbb{C}\setminus\{0\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq -1;\ \operatorname{ord}_\infty(f) \geq -1 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{matrix} f \text{ may have a pole of order} \leq 1 \text{ at } \infty \\ f \text{ may have a pole of order} \leq 1 \text{ at } 0 \end{matrix} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = \frac{P(z)}{z} \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 1 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(wP(1/w)\right) = 1 - \deg P \geq -1,$$

so $\deg P \leq 2$. Thus $P(z) = az^2 + bz + c$ $(a, b, c \in \mathbb{C})$ and therefore

$$f(z) = \frac{az^2+bz+c}{z} = az + b + \frac{c}{z}, \quad a, b, c \in \mathbb{C}.$$

Hence

$$L([0]+[\infty]) = \left\{ az + b + c/z : a, b, c \in \mathbb{C} \right\} = \operatorname{span}\left\{1, z, \frac{1}{z}\right\} \quad \text{and} \quad \ell([0]+[\infty]) = 3.$$

(7) Let $D = [\infty] - [1]$. Then

$$\begin{aligned} L([\infty]-[1]) &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [\infty] - [1] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p\in\mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f)-1)[0] + (\operatorname{ord}_\infty(f)+1)[\infty] \geq 0 \,\} \cup \{0\} \\ &= \{\, f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p\in\mathbb{C}\setminus\{0\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq 1;\ \operatorname{ord}_\infty(f) \geq -1 \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{matrix} f \text{ may have a pole of order at most 1 at } \infty \\ f \text{ has a zero of order at least 1 at 1} \end{matrix} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = (z-1)P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 1 at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(\frac{1-w}{w}\cdot P(1/w)\right) = -1 - \deg P \geq -1,$$

so $\deg P \leq 0$. Thus $P(z) = a$ $(a \in \mathbb{C})$ and therefore

$$f(z) = a(z-1), \quad a \in \mathbb{C}.$$

Hence

$$L([\infty]-[1]) = \{a(z-1) : a \in \mathbb{C}\} = \mathrm{span}_{\mathbb{C}}\{z-1\} \quad \text{and} \quad \ell([\infty]-[1]) = 1.$$

(8) Let $D = 2[\infty]-[1]$. Then

$$L(2[\infty]-[1]) = \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order at most 2 at } \infty \\ f \text{ has a zero of order at least 1 at 1} \end{array} \right\} \cup \{0\}$$

Let

$$f(z) = (z-1)P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 2 at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(\frac{1-w}{w}\cdot P(1/w)\right) = -1 - \deg P \geq -2,$$

so $\deg P \leq 1$. Thus $P(z) = az + b$ $(a, b \in \mathbb{C})$ and therefore

$$f(z) = (z-1)(az+b) = az^2 + (-a+b)z + b, \quad a, b \in \mathbb{C}.$$

Then

$$L(2[\infty]-[1]) \subseteq \mathrm{span}_{\mathbb{C}}\{1, z, z^2\} \quad \text{but} \quad L(2[\infty]-1) \neq \mathrm{span}_{\mathbb{C}}\{1, z, z^2\}.$$

since $L(2[\infty]-[1])$ has the condition that the function must must be vanish at $z = 1$. Thus,

$$f(z) = (z-1)(az+b) = az(z-1) + b(z-1), \quad a, b \in \mathbb{C}.$$

Hence

$$L(2[\infty]-[1]) = \{az(z-1) + b(z-1) : a, b \in \mathbb{C}\} = \mathrm{span}_{\mathbb{C}}\{z-1, z(z-1)\} \quad \text{and} \quad \ell(2[\infty]-[1]) = 2.$$

(9) Let $D = 2[\infty] - [0] - [1]$. Then

$$
\begin{aligned}
&L(2[\infty] - [0] - [1]) \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + 2[\infty] - [0] - [1] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0,1\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f) - 1)[0] + (\operatorname{ord}_1(f) - 1)[1] + (\operatorname{ord}_\infty(f) + 2)[\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \in \mathbb{C}\setminus\{0,1\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq 1;\ \operatorname{ord}_1(f) \geq 1;\ \operatorname{ord}_\infty(f) \geq -2 \} \cup \{0\} \\
&= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order at most 2 at } \infty \\ f \text{ has a zero of order at least 1 at } 0 \\ f \text{ has a zero of order at least 1 at } 1 \end{array} \right\} \cup \{0\}
\end{aligned}
$$

Let

$$f(z) = z(z-1)P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most 2 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(\frac{1}{w} \cdot \frac{1-w}{w} \cdot P(1/w) \right) = -2 - \deg P \geq -2,$$

so $\deg P \leq 0$. Thus $P(z) = a$ $(a \in \mathbb{C})$ and therefore $f(z) = az(z-1)$ $(a \in \mathbb{C})$. Hence

$$L(2[\infty] - [0] - [1]) = \{az(z-1) : a \in \mathbb{C}\} = \operatorname{span}_{\mathbb{C}}\{z(z-1)\} \quad \text{and} \quad \ell(2[\infty] - [0] - [1]) = 1.$$

(10) Let $D = [0] - [\infty]$. Then

$$
\begin{aligned}
L([0] - [\infty]) &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{div}(f) + [0] - [\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}\setminus\{0\}} \operatorname{ord}_p[p] + (\operatorname{ord}_0(f) + 1)[0] + (\operatorname{ord}_\infty(f) - 1)[\infty] \geq 0 \} \cup \{0\} \\
&= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \operatorname{ord}_{p \in \mathbb{C}\setminus\{0\}}(f) \geq 0;\ \operatorname{ord}_0(f) \geq -1;\ \operatorname{ord}_\infty(f) \geq 1 \} \cup \{0\} \\
&= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has a zero of order at least 1 at } \infty \\ f \text{ may have a pole of order at most 1 at } 0 \end{array} \right\} \cup \{0\}
\end{aligned}
$$

Let

$$f(z) = \frac{P(z)}{z} \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f has a zero of order at least 1 at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\left(w \cdot P(1/w) \right) = 1 - \deg P \geq 1,$$

so $\deg P \leq 0$. Thus $P(z) = a$ $(a \in \mathbb{C})$ and therefore $f(z) = a/z$ $(a \in \mathbb{C})$. Hence

$$L([0] - [\infty]) = \{a/z : a \in \mathbb{C}\} = \operatorname{span}_{\mathbb{C}}\left\{\frac{1}{z}\right\} \quad \text{and} \quad \ell([0] - [\infty]) = 1.$$

Note. Since every divisor on $\mathbb{CP}^1$ is equivalent to $n[\infty]$, it suffices to compute $L(n[\infty])$.

Proposition 4. *Let $D = n[\infty] \in \mathrm{Div}(\mathbb{CP}^1)$ for some $n \in \mathbb{Z}$. If $n \geq 0$, then*

$$L(n[\infty]) = \mathrm{span}_{\mathbb{C}} \left\{1, z, z^2, \ldots, z^n\right\}.$$

If $n < 0$, then $L(n[\infty]) = 0$. In particular,

$$\ell(n[\infty]) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

Proof. Take a divisor $D = n[\infty] \in \mathrm{Div}(\mathbb{CP}^1)$ with $n \in \mathbb{Z}$. Then

$$\begin{aligned} L(n[\infty]) &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{div}(f) + n[\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \textstyle\sum_{p \in \mathbb{C}} \mathrm{ord}_p[p] + (\mathrm{ord}_\infty(f) + n)[\infty] \geq 0 \} \cup \{0\} \\ &= \{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p \in \mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -n \} \cup \{0\}. \end{aligned}$$

(i) Assume that $n \geq 0$. Then

$$\begin{aligned} &\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p \in \mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -n \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ may have a pole of order at most } n \text{ at } \infty \\ f \text{ has no pole at } p \in \mathbb{C} \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f may have a pole of order at most n at ∞, we require

$$\mathrm{ord}_\infty(f) = \mathrm{ord}_{w=0}\left(P(1/w)\right) = -\deg P \geq -n,$$

so $\deg P \leq n$. Thus $P(z) = a$ $(a \in \mathbb{C})$ and therefore $f(z) = a_0 + a_1 z + \cdots + a_n z^n$ $(a_i \in \mathbb{C})$. Hence

$$L(n[\infty]) = \mathrm{span}_{\mathbb{C}} \left\{1, z, z^2, \ldots, z^n\right\} \quad \text{and} \quad \ell(n[\infty]) = n+1.$$

(ii) Assume that $n < 0$. Then

$$\begin{aligned} &\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \mathrm{ord}_{p \in \mathbb{C}}(f) \geq 0;\ \mathrm{ord}_\infty(f) \geq -n \} \cup \{0\} \\ &= \left\{ f \in \mathcal{M}(\mathbb{CP}^1)^\times \mid \begin{array}{c} f \text{ has a zero of order at least } -n \text{ at } \infty \\ f \text{ has no pole at } p \in \mathbb{C} \end{array} \right\} \cup \{0\} \end{aligned}$$

Let

$$f(z) = P(z) \quad \text{with} \quad P(z) \in \mathbb{C}[z].$$

Since f has a zero of order at least n at ∞, we require

$$\operatorname{ord}_\infty(f) = \operatorname{ord}_{w=0}\big(P(1/w)\big) = -\deg P \geq -n,$$

so $\deg P \leq n$. Since $P(z) \in \mathbb{C}[z]$, $P(z) \equiv 0$ therefore $f(z) = 0$. Hence

$$L(n[\infty]) = \{0\} \quad \text{and} \quad \ell(n[\infty]) = 0.$$

□

Theorem 5. *Let X be a compact Riemann surface, and let $D, D' \in \operatorname{Div}(X)$*

$$D \sim D' \quad \textit{implies} \quad L(D) \simeq L(D').$$

In particular, $\ell(D) = \ell(D')$.

Proof. Since $D \sim D'$,

$$\exists g \in \mathcal{M}(X)^\times \quad \text{such that} \quad D = D' + \operatorname{div}(g).$$

Define

$$\begin{array}{rccc} \Phi_g & : & L(D) & \longrightarrow & L(D') \\ & & f & \longmapsto & gf \end{array}.$$

Let $f \in L(D)$. Then $\operatorname{div}(f) + D \geq 0$, and so

$$\operatorname{div}\big(gf\big) + D' = (\operatorname{div}(g) + \operatorname{div}(f)) + (D - \operatorname{div}(g)) = \operatorname{div}(f) + D \geq 0.$$

Hence $gf \in L(D')$. So Φ_h is well-defined. Let f, f' and $a, b \in \mathbb{C}$ then

$$\Phi_g(af + bf') = g(af + bf') = agf + bgf' = a\Phi_g(f) + b\Phi_g(b),$$

so Φ_g be $\mathbb{C}$-linear. Define also

$$\begin{array}{rccc} \Psi_g & : & L(D) & \longrightarrow & L(D') \\ & & f & \longmapsto & \frac{f}{g} \end{array}.$$

We check that Ψ_h is well-defined. If $g \in L(D')$, then

$$\operatorname{div}(g) + D' \geq 0.$$

Since $D' = D + \operatorname{div}(h)$, we get

$$\operatorname{div}(gh) + D = \operatorname{div}(g) + \operatorname{div}(h) + D = \operatorname{div}(g) + D' \geq 0.$$

Thus

$$gh \in L(D),$$

so Ψ_h is well-defined.

Finally, for $f \in L(D)$ and $g \in L(D')$,

$$\Psi_h(\Phi_h(f)) = \left(\frac{f}{h}\right) h = f, \qquad \Phi_h(\Psi_h(g)) = \frac{gh}{h} = g.$$

Therefore Φ_h and Ψ_h are inverse linear maps. Hence

$$L(D) \cong L(D').$$

Taking dimensions, we conclude that

$$\ell(D) = \ell(D').$$

□

Corollary 6. *For any divisor D on $\mathbb{CP}^1$,*

$$\ell(D) = \begin{cases} \deg D + 1, & \deg D \geq 0, \\ 0, & \deg D < 0. \end{cases}$$

Remark 9. The condition is on $\deg D$, not on whether D is effective.

Proof. Note that $D \sim (\deg D)[\infty]$ on $\mathbb{CP}^1$. □

The correct formula on $\mathbb{CP}^1$ is

$$\ell(D) = \begin{cases} \deg D + 1, & \deg D \geq 0, \\ 0, & \deg D < 0. \end{cases}$$

The reason is that if $D \sim D'$, then $L(D) \cong L(D')$. Indeed, if

$$D - D' = \operatorname{div}(h)$$

for some nonzero meromorphic function h, then multiplication by h defines an isomorphism

$$L(D) \longrightarrow L(D'), \qquad f \longmapsto hf.$$

For if $f \in L(D)$, then

$$\operatorname{div}(f) + D \geq 0,$$

and therefore

$$\operatorname{div}(hf) + D' = \operatorname{div}(h) + \operatorname{div}(f) + D' = (D - D') + \operatorname{div}(f) + D' = \operatorname{div}(f) + D \geq 0.$$

Hence $hf \in L(D')$, and so

$$\ell(D) = \ell(D').$$

Now on $\mathbb{CP}^1$, every divisor D is linearly equivalent to $(\deg D)[\infty]$. Indeed, for every finite point $a \in \mathbb{C}$,

$$\operatorname{div}(z - a) = [a] - [\infty],$$

so

$$[a] \sim [\infty].$$

Therefore any divisor D satisfies

$$D \sim (\deg D)[\infty].$$

It follows that

$$\ell(D) = \ell((\deg D)[\infty]).$$

Since

$$\ell(n[\infty]) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0, \end{cases}$$

we conclude that

$$\ell(D) = \begin{cases} \deg D + 1, & \deg D \geq 0, \\ 0, & \deg D < 0. \end{cases}$$

The identity

$$L(D) = H^0(X, \mathcal{O}_X(D))$$

comes from the definition of the sheaf $\mathcal{O}_X(D)$. If

$$D = \sum_p n_p[p],$$

then for an open set $U \subseteq X$ one defines

$$\mathcal{O}_X(D)(U) = \{\, f \in \mathcal{M}_X(U) \mid \operatorname{div}(f)|_U + D|_U \geq 0 \,\},$$

where $\mathcal{M}_X(U)$ denotes the meromorphic functions on U. Thus a section of $\mathcal{O}_X(D)$ over U is precisely a meromorphic function whose poles and zeros are compatible with the divisor D on U.

Taking global sections gives

$$H^0(X, \mathcal{O}_X(D)) = \{\, f \in \mathcal{M}(X) \mid \operatorname{div}(f) + D \geq 0 \,\},$$

which is exactly the Riemann–Roch space $L(D)$.

Now let $X = \mathbb{CP}^1$ and $D = n[\infty]$. Then

$$H^0(\mathbb{CP}^1, \mathcal{O}(n[\infty])) = L(n[\infty]).$$

A function f lies in $L(n[\infty])$ if and only if it has no poles on $\mathbb{C}$ and has a pole of order at most n at ∞. Therefore f must be a polynomial of degree at most n. Hence

$$H^0(\mathbb{CP}^1, \mathcal{O}(n[\infty])) = L(n[\infty]) = \operatorname{span}\{1, z, z^2, \ldots, z^n\} \qquad (n \geq 0).$$

Using the local coordinate $w = 1/z$ near ∞, one has

$$z^k = w^{-k},$$

so z^k has a pole of order k at ∞. This explains why exactly the functions

$$1, z, \ldots, z^n$$

occur as global sections of $\mathcal{O}(n[\infty])$.

For now, you can think of

$$H^0(X, \mathcal{F})$$

as meaning simply the space of **global sections** of the sheaf $\mathcal{F}$. In other words,

$$H^0(X, \mathcal{F}) = \mathcal{F}(X).$$

So if $\mathcal{O}_X(D)$ is the sheaf whose sections over an open set U are meromorphic functions on U compatible with the divisor D, then

$$H^0(X, \mathcal{O}_X(D))$$

means the meromorphic functions on all of X compatible with D.

Thus

$$H^0(X, \mathcal{O}_X(D)) = \{ f \in \mathcal{M}(X) \mid \operatorname{div}(f) + D \geq 0 \} = L(D).$$

So the identity

$$L(D) = H^0(X, \mathcal{O}_X(D))$$

is just saying that the Riemann–Roch space $L(D)$ is the space of global sections of $\mathcal{O}_X(D)$.

For example, on $\mathbb{CP}^1$,

$$H^0(\mathbb{CP}^1, \mathcal{O}_{\mathbb{CP}^1}(n[\infty])) = L(n[\infty]) = \operatorname{span}\{1, z, z^2, \ldots, z^n\}.$$

Yes, but only as an analogy. The notation

$$H^0(X, \mathcal{O}_X(D))$$

is not de Rham cohomology; it is sheaf cohomology. However, there is a useful comparison.

In de Rham theory,

$$H^0_{\mathrm{dR}}(X) = \ker\big(d : \Omega^0(X) \to \Omega^1(X)\big),$$

so it is the space of global smooth functions f satisfying

$$df = 0.$$

Thus $H^0_{\mathrm{dR}}(X)$ consists of locally constant functions.

By contrast,

$$H^0(X, \mathcal{O}_X(D))$$

means the space of global sections of the sheaf $\mathcal{O}_X(D)$, that is,

$$H^0(X, \mathcal{O}_X(D)) = \mathcal{O}_X(D)(X).$$

Since $\mathcal{O}_X(D)$ is the sheaf of meromorphic functions compatible with the divisor D, one gets

$$H^0(X, \mathcal{O}_X(D)) = \{\, f \in \mathcal{M}(X) \mid \mathrm{div}(f) + D \geq 0 \,\} = L(D).$$

On a complex curve, the closest analytic analogue is not de Rham cohomology but the Dolbeault point of view:

$$H^0(X, \mathcal{O}_X(D)) = \ker\big(\bar{\partial} : \mathcal{A}^0(X, \mathcal{O}_X(D)) \to \mathcal{A}^{0,1}(X, \mathcal{O}_X(D))\big).$$

So one may think of $H^0(X, \mathcal{O}_X(D))$ as the space of global smooth sections of the line bundle $\mathcal{O}_X(D)$ that satisfy the holomorphicity condition $\bar{\partial} s = 0$.

For example, when $X = \mathbb{CP}^1$ and $D = n[\infty]$, one has

$$H^0(\mathbb{CP}^1, \mathcal{O}(n[\infty])) = L(n[\infty]) = \mathrm{span}\{1, z, z^2, \ldots, z^n\}.$$

A useful bridge is that both

$$H^0_{\mathrm{dR}}(X) \qquad \text{and} \qquad H^0(X, \mathcal{O}_X)$$

describe degree-zero global objects, but with different notions of "no variation."

In de Rham theory,

$$H^0_{\mathrm{dR}}(X) = \ker\bigl(d : \Omega^0(X) \to \Omega^1(X)\bigr).$$

Since $\Omega^0(X)$ is the space of smooth functions, this means

$$H^0_{\mathrm{dR}}(X) = \{\, f \in C^\infty(X) \mid df = 0 \,\}.$$

Thus $H^0_{\mathrm{dR}}(X)$ consists of locally constant functions, so if X is connected then

$$H^0_{\mathrm{dR}}(X) \cong \mathbb{C}.$$

On the holomorphic side,

$$H^0(X, \mathcal{O}_X) = \mathcal{O}_X(X)$$

is the space of global holomorphic functions on X. If X is a compact connected Riemann surface, every holomorphic function is constant, so

$$H^0(X, \mathcal{O}_X) = \mathbb{C}.$$

Thus on a compact connected Riemann surface, both degree-zero theories give constants:

$$H^0_{\mathrm{dR}}(X) \cong \mathbb{C}, \qquad H^0(X, \mathcal{O}_X) = \mathbb{C}.$$

The relation to the Riemann–Roch space is that

$$L(D) = H^0(X, \mathcal{O}_X(D)).$$

This is the same idea, except that instead of ordinary holomorphic functions one now allows meromorphic functions whose poles are controlled by the divisor D.

For example, on $\mathbb{CP}^1$,

$$H^0(\mathbb{CP}^1, \mathcal{O}) = \mathbb{C},$$

while for $D = n[\infty]$ one has

$$H^0(\mathbb{CP}^1, \mathcal{O}(n[\infty])) = L(n[\infty]) = \mathrm{span}\{1, z, z^2, \ldots, z^n\}.$$

So the sheaf $\mathcal{O}_X(D)$ enlarges the space of global sections by allowing controlled poles.

Consider the circle

$$S^1 = \{e^{it} : t \in \mathbb{R}/2\pi\mathbb{Z}\}.$$

A standard closed 1-form on S^1 is the local form

$$\omega = d\theta,$$

where θ is the angle coordinate. This form is closed, since locally

$$d\omega = d^2\theta = 0.$$

However, it is not globally exact, because

$$\int_{S^1} \omega = 2\pi \neq 0.$$

If $\omega = df$ globally, then by Stokes' theorem one would have

$$\int_{S^1} \omega = \int_{S^1} df = 0,$$

a contradiction.

Now choose an open cover $S^1 = U_0 \cup U_1 \cup U_2$ by arcs, each small enough that one can choose a smooth angle function θ_i on U_i. Then

$$\omega|_{U_i} = d\theta_i.$$

On overlaps $U_i \cap U_j$, one has

$$d(\theta_i - \theta_j) = d\theta_i - d\theta_j = \omega - \omega = 0,$$

so $\theta_i - \theta_j$ is constant on each connected overlap. Write

$$c_{ij} := \theta_i - \theta_j \in \mathbb{R}.$$

These constants satisfy

$$c_{ij} + c_{jk} + c_{ki} = 0$$

on triple overlaps, since

$$(\theta_i - \theta_j) + (\theta_j - \theta_k) + (\theta_k - \theta_i) = 0.$$

Thus (c_{ij}) is a Čech 1-cocycle with values in the constant sheaf $\underline{\mathbb{R}}$.

If $\omega = df$ globally, then on each U_i one has

$$d(\theta_i - f) = 0,$$

so $\theta_i - f = a_i$ for some constant a_i. Hence

$$c_{ij} = \theta_i - \theta_j = a_i - a_j,$$

so the cocycle is a coboundary. Conversely, if the cocycle is a coboundary, then the corrected local primitives glue to a global primitive. Therefore,

$$[\omega]_{\mathrm{dR}} = 0 \quad \Longleftrightarrow \quad (c_{ij}) \text{ is a coboundary.}$$

This is the basic bridge between de Rham cohomology and sheaf cohomology: a closed form gives local primitives, and the obstruction to gluing them into a global primitive is measured by a Čech cohomology class.

The holomorphic analogue of the de Rham story is obtained by replacing d with $\bar{\partial}$.
A smooth complex-valued function f on a complex manifold is holomorphic if and only if

$$\bar{\partial} f = 0.$$

Thus

$$H^0(X, \mathcal{O}_X)$$

may be viewed analytically as the space of global smooth functions satisfying

$$\bar{\partial} f = 0.$$

This is the holomorphic analogue of

$$H^0_{\mathrm{dR}}(X) = \ker(d : \Omega^0(X) \to \Omega^1(X)).$$

For degree 1, the analogy is with gluing local solutions. Suppose $\{U_i\}$ is an open cover of X, and suppose

$$g_{ij} \in \mathcal{O}_X(U_i \cap U_j)$$

is a Čech 1-cocycle, so that

$$g_{ij} + g_{jk} + g_{ki} = 0$$

on triple overlaps. This cocycle measures the obstruction to finding holomorphic functions $f_i \in$

$\mathcal{O}_X(U_i)$ such that

$$f_i - f_j = g_{ij}$$

on overlaps. If such f_i exist, then the cocycle is a coboundary.

To relate this to calculus, choose a smooth partition of unity $\{\rho_k\}$ subordinate to the cover and define smooth functions

$$b_i := \sum_k \rho_k \, g_{ik}$$

on U_i. Then on overlaps,

$$b_i - b_j = g_{ij}.$$

Applying $\bar{\partial}$, one gets

$$\bar{\partial} b_i - \bar{\partial} b_j = \bar{\partial}(g_{ij}) = 0,$$

since each g_{ij} is holomorphic. Hence the local forms $\bar{\partial} b_i$ glue to a global $(0,1)$-form

$$\alpha \in \mathcal{A}^{0,1}(X)$$

with

$$\bar{\partial}\alpha = 0.$$

Moreover, the cocycle (g_{ij}) is a coboundary if and only if α is $\bar{\partial}$-exact. This gives the bridge

$$H^1(X, \mathcal{O}_X) \cong H^{0,1}_{\bar{\partial}}(X),$$

which is the degree-1 case of Dolbeault's theorem.

So the analogy is:

$$\text{de Rham: } d \qquad \leftrightarrow \qquad \text{Dolbeault: } \bar{\partial},$$

and

$$H^1_{\mathrm{dR}}(X)$$

measures failure to glue local primitives, while

$$H^1(X, \mathcal{O}_X)$$

measures failure to glue local holomorphic functions.

Remark 10. The space $L(D)$ is a vector space over a field $\mathbb{C}$ because it is a subspace of the field of meromorphic functions $\mathcal{M}(\mathbb{CP}^1)$. Its operations are the usual ones:

$$+ : L(D) \times L(D) \to L(D), \qquad (f, g) \mapsto f + g,$$

and

$$\cdot : \mathbb{C} \times L(D) \to L(D), \qquad (\alpha, f) \mapsto \alpha f.$$

To check that $L(D)$ is closed under scalar multiplication, let $f \in L(D)$ and $a \in F$. If $a = 0$, then $af = 0 \in L(D)$. If $a \neq 0$, then

$$\operatorname{div}(af) = \operatorname{div}(f),$$

so

$$\operatorname{div}(af) + D = \operatorname{div}(f) + D \geq 0.$$

Hence $af \in L(D)$.

To check closure under addition, let $f, g \in L(D)$. For every point p, one has

$$\operatorname{ord}_p(f + g) \geq \min\big(\operatorname{ord}_p(f), \operatorname{ord}_p(g)\big).$$

Therefore

$$\operatorname{ord}_p(f + g) + D(p) \geq \min\big(\operatorname{ord}_p(f) + D(p), \operatorname{ord}_p(g) + D(p)\big).$$

Since $f, g \in L(D)$, both terms on the right are nonnegative, so

$$\operatorname{ord}_p(f + g) + D(p) \geq 0$$

for every p. Thus

$$\operatorname{div}(f + g) + D \geq 0,$$

and hence $f + g \in L(D)$.

So $L(D)$ is a vector space over F.

Example 5. Take

$$P = [0:1], \qquad Q = [1:1], \qquad D = P - Q.$$

on $\mathbb{CP}^1$. Using the affine coordinate $z = \frac{X}{Y}$, we identify

$$P = [0:1] \leftrightarrow z = 0,$$
$$Q = [1:1] \leftrightarrow z = 1.$$

So in this chart, the divisor be $D = [0] - [1]$. Then

$$L(P-Q) = \left\{ f \in M(\mathbb{CP}^1)^\times : \operatorname{div}(f) + P - Q \geq 0 \right\} \cup \{0\}.$$

Since

$$\operatorname{div}(f) + P + Q = \sum_{R \in \mathbb{CP}^1} (\operatorname{ord}_R(f) + n_P - n_Q)[R] \quad \text{with} \quad n_P - n_Q = \begin{cases} 1 & R = P, \\ -1 & R = Q, \\ 0, & R \neq P, Q, \end{cases}$$

we have

$$\operatorname{div}(f) + P - Q \geq 0 \implies \begin{cases} \operatorname{ord}_P(f) + 1 \geq 0 & \text{at } P \\ \operatorname{ord}_Q(f) - 1 \geq 0 & \text{at } Q \\ \operatorname{ord}_R(f) \geq 0 & \text{at } R \neq P, Q \end{cases} = \begin{cases} \operatorname{ord}_P(f) \geq -1 & \text{at } P \\ \operatorname{ord}_Q(f) \geq 1 & \text{at } Q \\ \operatorname{ord}_R(f) \geq 0 & \text{at } R \neq P, Q \end{cases},$$

so

$$f \in L(P-Q) \iff \begin{cases} f \text{ may have a pole of order at most 1 at } P, \\ f \text{ has a zero of order at least 1 at } Q, \\ f \text{ has no poles at any point other than } P. \end{cases}$$

Suppose that

$$f \in L(P-Q).$$

To ensure that there are no poles anywhere on CP^1, define.

$$g(z) := \frac{z}{z-1} \cdot f(z).$$

Since $g = c \in \mathbb{C}$ be constant,

$$f(z) = c \cdot \frac{z-1}{z}.$$

Thus,

$$L(P-Q) = \mathbb{C} \cdot \frac{z-1}{z}$$

and $\ell(P-Q) = \dim_{\mathbb{C}} L(P-Q) = 1$.

Definition 6. Let $f \in \mathcal{M}(X)^\times$ be a nonzero meromorphic function on X. Its principal divisor is

$$(f) := \operatorname{div}(f) := \sum_{p \in X} \operatorname{ord}_p(f)\,[p].$$

The subgroup of principal divisors is

$$\operatorname{Prin}(X) := \{\, (f) \mid f \in \mathcal{M}(X)^\times \,\} \subset \operatorname{Div}(X).$$

Definition 7. Two divisors $D, E \in \operatorname{Div}(X)$ are said to be linearly equivalent if

$$D - E \in \operatorname{Prin}(X).$$

Equivalently, $D \sim E$ if and only if there exists $f \in \mathcal{M}(X)^\times$ such that

$$D - E = (f).$$

Definition 8. The Picard group of X is the quotient group

$$\operatorname{Pic}(X) := \operatorname{Div}(X)/\operatorname{Prin}(X).$$

Thus $\operatorname{Pic}(X)$ is the group of linear equivalence classes of divisors on X. The class of a divisor D in $\operatorname{Pic}(X)$ is denoted by $[D]$.

Example 6. Let $X = \mathbb{CP}^1$ and $D = n[\infty]$. Then

$$L(n[\infty]) = \{\, f \in \mathcal{M}(\mathbb{CP}^1) \mid \operatorname{div}(f) + n[\infty] \geq 0 \,\} \cup \{0\}.$$

This means that f has no poles on $\mathbb{C}$, and at ∞ it may have a pole of order at most n. Hence f is a polynomial of degree at most n. Therefore

$$L(n[\infty]) = \operatorname{span}_{\mathbb{C}}\{1, z, z^2, \ldots, z^n\} \qquad (n \geq 0).$$

One can interpret $L(n[\infty])$ using the logarithmic derivative

$$\omega_f := \frac{1}{2\pi i}\frac{df}{f},$$

whose integral over a loop computes a winding number. For a small positively oriented loop γ_p around a point p, one has

$$\mathrm{ord}_p(f) = \int_{\gamma_p} \omega_f.$$

Thus the condition

$$f \in L(n[\infty]) \qquad \Longleftrightarrow \qquad \mathrm{div}(f) + n[\infty] \geq 0$$

means that $\mathrm{ord}_p(f) \geq 0$ for every $p \neq \infty$, so f has no poles away from ∞, and that

$$\mathrm{ord}_\infty(f) \geq -n,$$

so f has a pole of order at most n at infinity.

If $X = \mathbb{P}^1$ with affine coordinate z, then using the local coordinate $w = 1/z$ near ∞ gives

$$\mathrm{ord}_\infty(f) = -\frac{1}{2\pi i}\int_{|z|=R} \frac{f'(z)}{f(z)}\,dz \qquad (R \gg 0).$$

Therefore

$$f \in L(n[\infty])$$

if and only if f is holomorphic on $\mathbb{C}$ and

$$\frac{1}{2\pi i}\int_{|z|=R} \frac{f'(z)}{f(z)}\,dz \leq n$$

for all sufficiently large R. In other words, the image of a large circle under f winds around 0 at most n times. On $\mathbb{P}^1$, this is exactly the behavior of a polynomial of degree at most n. Hence

$$L(n[\infty]) = \{\text{polynomials of degree at most } n\}.$$

Proposition 7. *For every divisor* $D \in \mathrm{Div}(\mathbb{CP}^1)$*, one has*

$$D \sim (\deg D)[\infty].$$

Proof. Let $a \in \mathbb{C} \subset \mathbb{CP}^1$. The meromorphic function $z - a$ on $\mathbb{CP}^1$ has a simple zero at $z = a$ and a simple pole at ∞. Hence

$$\mathrm{div}(z - a) = [a] - [\infty].$$

Therefore

$$[a] \sim [\infty]$$

for every finite point $a \in \mathbb{C}$.

Now let

$$D = \sum_{p \in \mathbb{CP}^1} n_p [p].$$

Since each point $[p]$ is linearly equivalent to $[\infty]$, it follows that

$$D \sim \sum_{p \in \mathbb{CP}^1} n_p [\infty].$$

But

$$\sum_{p \in \mathbb{CP}^1} n_p = \deg D,$$

so

$$\sum_{p \in \mathbb{CP}^1} n_p [\infty] = (\deg D)[\infty].$$

Thus

$$D \sim (\deg D)[\infty].$$

□

5 The Canonical Divisor

TBA

6 Riemann–Roch Theorem

TBA